
Adaptive Touchless Authentication System

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Abstract

Secure and intuitive user authentication has become increasingly important in modern human-computer interaction, particularly in applications requiring contactless operation and seamless user experience. Conventional authentication methods such as passwords, PINs, RFID cards, and fingerprint-based systems often suffer from limitations including physical interaction, credential theft, hardware dependency, and reduced usability in touch-free environments. This paper presents the design and implementation of an Adaptive Touchless Authentication System based on behavioral gesture biometrics, developed using a low-cost ESP32 microcontroller integrated with OpenCV, MediaPipe, UART communication, and Bluetooth Human Interface Device (HID) technology. The proposed system utilizes a standard laptop webcam to capture hand gestures and employs MediaPipe for real-time hand landmark detection. A sequence of predefined gestures is analyzed to extract behavioral features including gesture speed, movement duration, execution timing, and sequence consistency, which together form a unique user template during the registration phase. These extracted features are transmitted to the ESP32 through UART communication, where user templates are securely stored and compared using a similarity-based authentication algorithm. The ESP32 computes a similarity score between the stored template and the current gesture sequence to determine whether access should be granted based on a predefined authentication threshold. To further enhance system security, the proposed design incorporates multi-user registration, wrong-sequence lockout protection, and repeated authentication attempt monitoring. Upon successful authentication, the ESP32 functions as a Bluetooth HID keyboard and wirelessly controls Microsoft PowerPoint presentations through predefined gesture commands such as next slide, previous slide, start slideshow, and exit presentation. The proposed embedded architecture combines computer vision with behavioral biometrics and embedded decision making to provide a secure, low-cost, and contactless authentication solution suitable for smart classrooms, corporate presentations, interactive kiosks, healthcare environments, and other human-computer interaction applications.

Keywords: Touchless Authentication, Behavioral Biometrics, Gesture Recognition, OpenCV, MediaPipe, UART Communication, Bluetooth HID, Similarity Score.

1. Introduction

In an increasingly digital and interconnected world, secure and intuitive user authentication has become an essential requirement across educational institutions, corporate environments, healthcare facilities, and smart interactive systems. Conventional authentication techniques such as passwords, Personal Identification Numbers (PINs), RFID cards, and fingerprint-based verification have been widely adopted

due to their simplicity and accessibility. However, these methods suffer from several limitations including physical interaction, password theft, forgotten credentials, sensor dependency, and vulnerability to unauthorized access. Furthermore, the growing demand for contactless human-computer interaction has accelerated the need for authentication systems that are both secure and convenient without requiring specialized biometric hardware.

The core motivation behind this project arises from the observation that most existing gesture-controlled presentation systems focus only on recognizing predefined hand gestures for executing commands, while completely overlooking user authentication. In such systems, any individual capable of reproducing the required gesture can operate the presentation, creating a significant security risk in environments where presentation access should be restricted to authorized users. Similarly, many gesture recognition systems rely entirely on software-based processing running on personal computers and do not leverage embedded hardware for secure decision making, reducing system portability and limiting practical deployment.

To address these challenges, the proposed Adaptive Touchless Authentication System integrates computer vision with embedded authentication by combining OpenCV, MediaPipe, and an ESP32 microcontroller. The system utilizes a standard laptop webcam to capture hand gestures and employs MediaPipe to detect hand landmarks in real time. Instead of relying solely on gesture recognition, the system extracts multiple behavioral biometric features including gesture sequence, movement speed, execution duration, timing, and motion consistency. During the registration phase, these features are used to create a unique user template that is securely stored within the ESP32. During authentication, the extracted behavioral features are transmitted to the ESP32 through UART communication, where a similarity-based matching algorithm compares the incoming gesture data with the stored templates to determine user authenticity.

To further strengthen system security, the proposed architecture incorporates multi-user registration, wrong-sequence lockout protection, configurable authentication thresholds, and repeated authentication attempt monitoring. These mechanisms enable the system to distinguish legitimate users from unauthorized individuals while minimizing false acceptance and false rejection rates. Upon successful authentication, the ESP32 functions as a Bluetooth Human Interface Device (HID), wirelessly emulating a keyboard to control Microsoft PowerPoint presentations using predefined hand gestures such as next slide, previous slide, start slideshow, and exit presentation. This dual functionality transforms the ESP32 into both an embedded authentication engine and a wireless presentation controller.

2. Literature Survey

Gesture-based human-computer interaction and behavioral biometric authentication have emerged as significant research areas due to the increasing demand for secure, contactless, and intuitive interaction with digital systems. Recent advancements in computer vision, artificial intelligence, embedded systems, and wireless communication have enabled the development of intelligent authentication systems capable of recognizing user behavior without requiring physical contact. Numerous studies have investigated various techniques for hand gesture recognition, biometric authentication, feature extraction, and embedded implementation. A comprehensive review of these existing approaches provides valuable insight into their

advantages and limitations while identifying the research gap addressed by the proposed Adaptive Touchless Authentication System.

Early gesture recognition systems primarily relied on colored gloves, depth sensors, or specialized hardware to detect hand movements. Although these approaches achieved satisfactory recognition accuracy, they significantly increased system cost and reduced portability. Furthermore, many of these systems required controlled environments and dedicated sensing hardware, limiting their practical deployment in everyday applications.

Freeman and Roth [1] introduced one of the earliest vision-based hand gesture recognition techniques using orientation histograms extracted from hand images. Their work demonstrated that vision-based gesture recognition could eliminate the need for wearable devices and enable natural human-computer interaction. However, the proposed approach was highly dependent on image quality and background conditions, making it sensitive to illumination variations and complex scenes. Such limitations reduce its reliability in real-time applications where environmental conditions frequently change.

OpenCV Contributors and Bradski [2] developed the Open Source Computer Vision library, providing a comprehensive framework for image acquisition, preprocessing, feature extraction, and real-time object detection. OpenCV significantly simplified the implementation of vision-based applications and became one of the most widely adopted computer vision platforms. Although OpenCV offers powerful image processing capabilities, it does not inherently provide accurate hand landmark detection or behavioral biometric analysis. Developers are therefore required to integrate additional machine learning frameworks for robust gesture recognition.

Lugaresi et al. [3] introduced MediaPipe Hands, a machine learning framework capable of detecting twenty-one three-dimensional hand landmarks in real time using a single RGB camera. The framework demonstrated high detection accuracy while maintaining low computational requirements, making it suitable for real-time human-computer interaction. However, MediaPipe focuses primarily on spatial landmark estimation and gesture recognition rather than user authentication. It does not consider behavioral characteristics such as gesture execution speed, movement timing, or motion consistency, allowing different individuals to reproduce identical gesture sequences without user verification.

Jain, Ross, and Pankanti [4] extensively investigated behavioral biometric authentication techniques, emphasizing that human behavioral characteristics provide an additional security layer beyond conventional physiological biometrics. Their research demonstrated that behavioral traits such as movement dynamics, execution timing, and interaction patterns exhibit unique characteristics for individual users. Although behavioral biometrics significantly improve authentication reliability, their implementation often requires sophisticated feature extraction algorithms and computationally intensive matching techniques, limiting their applicability in lightweight embedded systems.

Espressif Systems [5] introduced the ESP32 microcontroller as an integrated embedded platform supporting Wi-Fi, Bluetooth, non-volatile storage, UART communication, and real-time peripheral interfaces within a low-cost dual-core architecture. The ESP32 has become widely adopted for Internet of Things (IoT) and embedded automation applications due to its processing capability and wireless

communication features. Nevertheless, most existing implementations utilize the ESP32 primarily for data acquisition or communication, while complex authentication algorithms continue to execute on external computers. Consequently, the embedded processing capabilities of the ESP32 remain underutilized in behavioral authentication applications.

The proposed Adaptive Touchless Authentication System addresses these research gaps by integrating computer vision with embedded behavioral authentication. OpenCV and MediaPipe are employed to perform real-time hand landmark detection and gesture recognition, while behavioral features including gesture sequence, execution speed, duration, timing, and consistency are extracted to create a unique user template. Instead of performing authentication entirely on the laptop, these features are transmitted to an ESP32 through UART communication, where user templates are securely stored and compared using a similarity-based authentication algorithm. The ESP32 performs multi-user matching, configurable threshold verification, and wrong-sequence lockout before making the final access decision. Following successful authentication, the ESP32 operates as a Bluetooth Human Interface Device (HID) to wirelessly control Microsoft PowerPoint presentations. This hybrid architecture combines the advantages of computer vision, behavioral biometrics, embedded processing, and wireless communication while maintaining low computational complexity, reduced system cost, and improved security for real-time touchless authentication applications.

Ref	Authors	Method	Environment	Key Results
[1]	Freeman & Roth	Vision-based Hand Gesture Recognition using Orientation Histograms	Indoor Human-Computer Interaction	Demonstrated natural gesture recognition without wearable devices; performance affected by lighting and complex backgrounds.
[2]	Gary Bradski & OpenCV Contributors	OpenCV Computer Vision Framework	General Computer Vision Applications	Provided efficient real-time image processing tools but lacked built-in behavioral authentication capabilities.
[3]	Lugaresi et al.	MediaPipe Hands using Machine Learning-based Hand Landmark Detectionlogic	Real-Time RGB Camera Applications	Accurate real-time hand landmark detection with low computational cost; did not perform user authentication.
[4]	Jain, Ross & Pankanti	Behavioral Biometric Authentication	Biometric Security Systems	Demonstrated uniqueness of behavioral characteristics for authentication; required complex feature extraction and matching algorithms.
[5]	Espressif Systems	ESP32 Embedded Platform with Bluetooth andUART Communication	Embedded IoT and Automation Systems	Low-cost embedded platform supporting wireless communication; authentication logic generally executed on external computers rather than on-device.

Table 1: Summary of Existing Gesture Recognition, Behavioral Biometric Authentication Systems

3. Proposed Algorithm

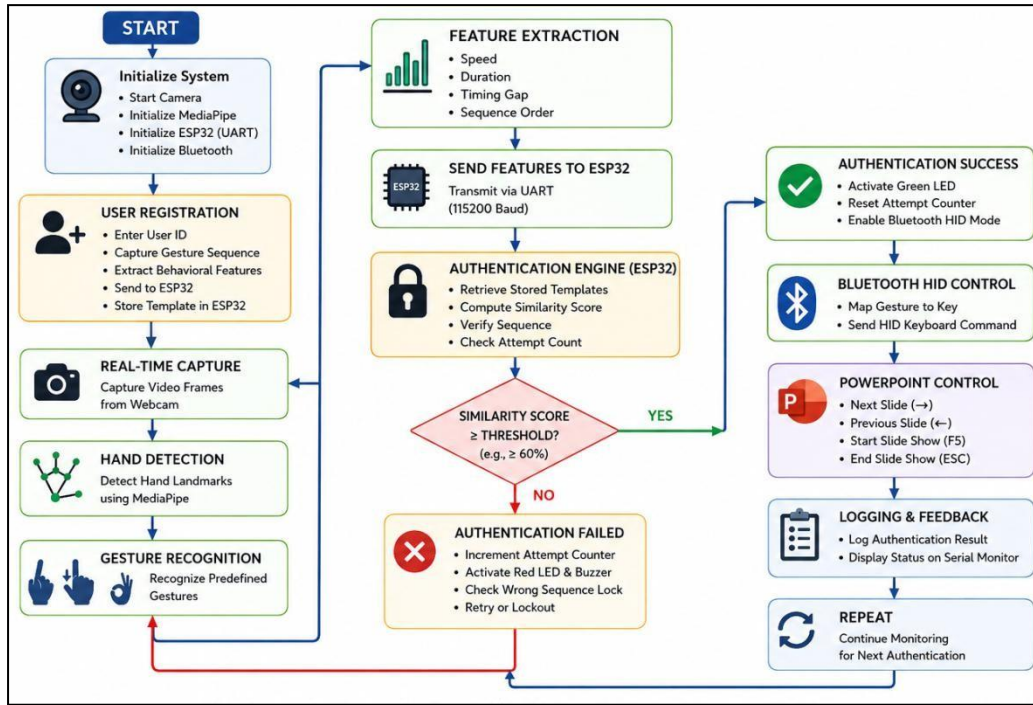
The proposed Adaptive Touchless Authentication System integrates computer vision, behavioral biometrics, embedded authentication, and Bluetooth Human Interface Device (HID) communication to provide secure and contactless user verification. The system operates in two distinct phases: Registration and Authentication. During the registration phase, behavioral gesture features are extracted and stored as user templates in the ESP32. During authentication, the current gesture is compared with the stored template using a similarity-based matching algorithm. Only authenticated users are permitted to control the PowerPoint presentation through Bluetooth HID.

The complete procedure is summarized in Algorithm 1.

Algorithm 1 Adaptive Touchless Authentication and Presentation Control

Require: Webcam, OpenCV, MediaPipe, ESP32.

- 1: Initialize webcam, OpenCV, MediaPipe, UART communication, and ESP32
- 2: Load registered user templates from ESP32 non-volatile memory
- 3: Capture continuous video frames from the webcam
- 4: Detect hand landmarks using the MediaPipe Hands model
- 5: Recognize the performed gesture sequence
- 6: Extract behavioral features
 - Gesture sequence
 - Movement speed
 - Gesture timing
- 7: If the system is in Registration Mode, then
- 8: Generate a behavioral gesture template
- 9: Transmit the template to the ESP32 via UART
- 10: Store the template in non-volatile memory
- 11: Else (Authentication Mode)
- 12: Transmit the extracted feature vector to the ESP32
- 13: Retrieve all registered user templates
- 14: Compute similarity scores for each template
- 15: Identify the highest matching user
- 16: If the similarity score $\geq$ authentication threshold and the gesture sequence is valid, then
- 17: Grant authentication
- 18: Convert subsequent gestures into PowerPoint commands
- 19: Else
- 20: Deny authentication
- 21: End If
- 22: Continue real-time gesture acquisition until system termination



Flowchart of the Proposed Adaptive Authentication Algorithm

4. System Implementation Architecture

The implementation architecture of the proposed Adaptive Touchless Authentication System integrates computer vision, behavioral biometrics, embedded authentication, and wireless communication within a hybrid hardware–software framework. The system is designed around two major processing units: a laptop, which performs real-time computer vision and gesture feature extraction, and an ESP32 microcontroller, which serves as the embedded authentication engine and Bluetooth Human Interface Device controller.

The laptop utilizes a standard webcam to continuously capture live video of the user's hand gestures. The captured frames are processed using the OpenCV computer vision library together with the MediaPipe Hands framework, which accurately detects twenty-one three-dimensional hand landmarks in real time. Based on these landmarks, the system recognizes predefined gesture sequences and extracts behavioral biometric features including gesture sequence, movement speed, execution duration, gesture timing, and motion consistency. These behavioral characteristics form a unique feature vector representing the user's gesture dynamics.

The extracted feature vector is transmitted to the ESP32 through UART serial communication, enabling efficient and low-latency data exchange between the laptop and the embedded controller. The ESP32 receives the behavioral features and performs all authentication-related computations independently of the laptop. During the registration phase, the ESP32 stores user templates in its non-volatile memory, allowing multiple users to be enrolled without requiring external storage. During authentication, the ESP32 retrieves the stored templates and computes similarity scores between the current behavioral feature vector and each registered template. The user is authenticated only when both the gesture sequence and the similarity score

satisfy the predefined authentication threshold. To further improve security, the system incorporates multi-user authentication, wrong-sequence lockout protection, and repeated authentication attempt monitoring to prevent unauthorized access.

Following successful authentication, the ESP32 automatically switches from authentication mode to Bluetooth Human Interface Device mode. In this mode, the microcontroller emulates a standard wireless Bluetooth keyboard and converts recognized gestures into predefined keyboard commands. These commands are transmitted wirelessly to the connected computer, enabling hands-free control of Microsoft PowerPoint presentations. Typical gesture mappings include advancing to the next slide, returning to the previous slide, starting the slideshow, and exiting presentation mode. If authentication fails, Bluetooth HID functionality remains disabled, ensuring that unauthorized users cannot operate the presentation.

Hardware Components

ESP32 Board:

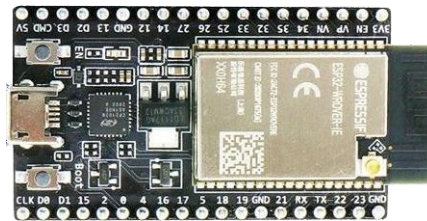


Fig 1: ESP32 Board

The ESP32 serves as the central embedded controller responsible for receiving behavioral gesture features through UART communication, storing registered user templates in non-volatile memory, calculating similarity scores, performing authentication decisions, and functioning as a Bluetooth HID keyboard for wireless PowerPoint presentation control.

Webcam:



Fig 2: Webcam

A standard USB webcam captures live video frames of the user's hand gestures. The webcam serves as the image acquisition device and eliminates the need for specialized depth cameras or dedicated gesture sensing hardware.

Software Implementation

The software architecture consists of two independent yet interconnected modules: a Laptop-Based Computer Vision Module and an Embedded Authentication Module.

Laptop Software:

The laptop application is developed using Python and integrates the following software libraries:

- OpenCV for real-time image acquisition and image processing

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- MediaPipe Hands for hand landmark detection and gesture recognition
 - PySerial for UART communication with the ESP32

The laptop continuously captures video frames, detects hand landmarks, recognizes gesture sequences, extracts behavioral biometric features, and transmits the generated feature vectors to the ESP32.

Embedded Software:

The embedded firmware is developed using ESP-IDF in the C programming language. The firmware is organized into modular source files to improve maintainability and scalability.

5. Working of the Proposed System

The proposed Adaptive Touchless Authentication System operates through a sequential process involving system initialization, user registration, behavioral gesture authentication, similarity-based decision making, and wireless presentation control. The overall workflow combines computer vision with embedded processing to provide secure and contactless access to Microsoft PowerPoint presentation control.

A. System Initialization

At startup, the laptop initializes the webcam, OpenCV library, MediaPipe Hands framework, and UART serial communication. Simultaneously, the ESP32 initializes its UART interface, Bluetooth Human Interface Device module, and Non-Volatile Storage, where registered user templates are stored. Once initialization is complete, a communication link is established between the laptop and the ESP32, allowing continuous exchange of behavioral gesture data.

B. User Registration

Before authentication, each user performs a predefined sequence of hand gestures in front of the webcam. The webcam continuously captures video frames, and MediaPipe detects twenty-one hand landmarks in real time. OpenCV recognizes the gesture sequence and extracts behavioral biometric features including gesture sequence, movement speed, execution duration, gesture timing, and motion consistency.

The extracted feature vector is transmitted to the ESP32 through UART communication, where it is securely stored in Non-Volatile Storage as the user's authentication template. Multiple users can be registered independently, enabling the system to support multi-user authentication.

C. Gesture Acquisition and Feature Extraction

During authentication, the webcam continuously captures live video frames while MediaPipe tracks the user's hand movements. OpenCV processes the detected landmarks to recognize predefined gestures and extract the associated behavioral biometric features. Instead of transmitting complete image frames, only the extracted feature vector is sent to the ESP32, significantly reducing communication overhead and improving system efficiency.

D. Authentication and Similarity Matching

The ESP32 receives the behavioral feature vector through UART communication and retrieves all previously registered user templates from memory. A similarity matching algorithm compares the current

gesture with each stored template based on gesture sequence, movement speed, execution duration, timing, and consistency.

The template with the highest similarity score is selected as the best match. Authentication is granted only if the similarity score exceeds the predefined threshold and the detected gesture sequence matches the registered template. If the similarity score falls below the threshold or an incorrect gesture sequence is detected, authentication is denied. The system also monitors repeated failed authentication attempts and activates a temporary lockout mechanism to prevent unauthorized access.

E. Bluetooth Presentation Control

Upon successful authentication, the ESP32 automatically switches to Bluetooth HID mode and functions as a wireless keyboard. Subsequent hand gestures are interpreted as presentation control commands and transmitted to the connected computer through Bluetooth.

The gesture-to-command mapping is defined as follows:

Right Swipe → Next Slide	Up Swipe → Start Presentation
Left Swipe → Previous Slide	Down Swipe → Exit Presentation

6. Experimental Results

The proposed Adaptive Touchless Authentication System was experimentally evaluated under various real-world operating conditions to assess its authentication accuracy, response time, reliability, and overall usability. The experimental setup consisted of a laptop equipped with a standard USB webcam running OpenCV and MediaPipe for real-time gesture recognition, connected to an ESP32 microcontroller through UART communication. The ESP32 was configured to perform behavioral biometric authentication, similarity score calculation, user template matching, and Bluetooth Human Interface Device communication for Microsoft PowerPoint presentation control.

A series of authentication and presentation control experiments were conducted in indoor environments under different lighting conditions and with multiple users. Approximately 20–25 test cases were performed, including successful authentication, unauthorized user attempts, incorrect gesture sequences, repeated authentication failures, and PowerPoint presentation control operations. The objective of these experiments was to evaluate the robustness of the behavioral authentication algorithm, verify real-time communication between the laptop and ESP32, and validate reliable wireless presentation control.

Experimental observations demonstrated that the proposed system successfully distinguished between registered and unregistered users by evaluating both gesture sequence and behavioral characteristics. Genuine users consistently produced high similarity scores exceeding the predefined authentication threshold, while unauthorized users or incorrect gesture sequences resulted in significantly lower similarity scores, leading to authentication denial. The implemented wrong-sequence lockout mechanism further enhanced system security by preventing repeated unauthorized authentication attempts.

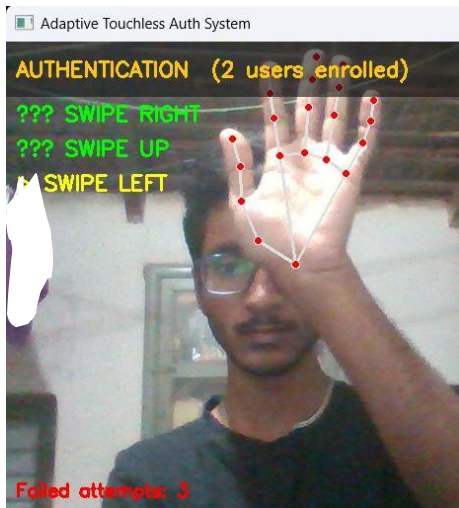


Fig 3: Camera detecting a gesture in attempt 4

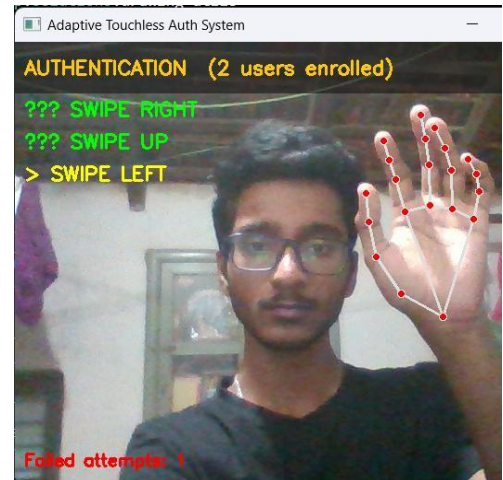


Fig 4: Camera detecting a gesture in attempt 2

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Checking against all users...
mm: 25.3/100
nischal: 65.7/100

[✓] ACCESS GRANTED
Welcome, NISCHAL! (65.7/100)
Opening PowerPoint...
PowerPoint slideshow started!
Camera window hidden.

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Fig 5: Serial Monitor displaying successful authentication.

Following successful authentication, the ESP32 automatically enabled Bluetooth HID mode and functioned as a wireless keyboard. Presentation control commands generated through predefined gestures were successfully transmitted to Microsoft PowerPoint with minimal communication delay. Right and left swipe gestures reliably navigated between presentation slides, while upward and downward gestures successfully initiated and terminated slideshow mode. The transition between authentication mode and presentation control mode occurred seamlessly, ensuring secure operation without noticeable latency.

7. Conclusion

This work presents an Adaptive Touchless Authentication System that combines computer vision, behavioral biometrics, embedded authentication, and wireless Human Interface Device communication to provide a secure and contactless user authentication solution. By integrating OpenCV and MediaPipe with an ESP32 microcontroller, the proposed system successfully performs real-time gesture recognition, behavioral feature extraction, similarity-based authentication, and wireless PowerPoint presentation control. Unlike conventional gesture-controlled presentation systems that directly execute commands upon gesture recognition, the proposed architecture introduces an embedded authentication layer that verifies the user's identity before enabling presentation control, thereby significantly enhancing system security.

A major contribution of this project is the implementation of behavioral biometric authentication using gesture dynamics rather than relying solely on gesture recognition. The system evaluates multiple behavioral characteristics, including gesture sequence, movement speed, execution duration, timing, and motion consistency, to generate a unique user template for each registered individual. These behavioral features are securely stored within the ESP32 and compared with real-time gesture data using a similarity-based matching algorithm. This approach enables the system to distinguish between authorized and unauthorized users while reducing the possibility of unauthorized presentation access through simple gesture imitation.

Another significant contribution of the proposed system is the distribution of computational tasks between the laptop and the embedded controller. Computationally intensive computer vision operations, including hand landmark detection and feature extraction, are executed on the laptop using OpenCV and MediaPipe, while authentication, similarity score computation, template management, and access decision making are performed directly on the ESP32. This hybrid architecture efficiently utilizes the strengths of both platforms, resulting in low computational complexity, real-time performance, improved scalability, and enhanced embedded security without requiring high-end hardware or expensive biometric sensors.

The integration of Bluetooth HID functionality further extends the capabilities of the proposed system by enabling the ESP32 to function as a wireless keyboard after successful authentication. Once access is granted, predefined hand gestures are seamlessly translated into PowerPoint presentation commands, allowing users to navigate slides without physical interaction with conventional input devices. Experimental evaluation demonstrated reliable gesture recognition, accurate similarity-based authentication, low communication latency, and stable wireless presentation control under normal indoor operating conditions. The implemented multi-user registration, configurable authentication threshold, and wrong-sequence lockout mechanism further improve the overall robustness and security of the system.

Overall, the proposed Adaptive Touchless Authentication System effectively bridges the gap between traditional gesture-controlled interfaces and secure embedded authentication systems by integrating behavioral biometrics with real-time computer vision and embedded decision making. The developed architecture provides a low-cost, scalable, reliable, and practical solution for secure touchless human-computer interaction. Owing to its flexibility and modular design, the proposed system can be extended to applications such as smart classrooms, corporate meeting rooms, secure workstation login, interactive kiosks, healthcare environments, access control systems, and future intelligent embedded authentication platforms.

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